

TABLE 3-11
TABULATED REACTOR DATA

Initial Core Loading (<i>circa 1972</i>)	25 assemblies per core, 12.59 kg U-235
Present Core Loading	25 assemblies per core, 12.59 kg U-235, 5 graphite reflectors, 5 beryllium reflectors
Design Steady-State Power	1 MW _{TH}
<u>Fuel</u>	Pin type, 4% enriched pellets with Zircaloy-2 cladding
Material	UO ₂ - Sintered pellets
Enrichment	4.0 w% U-235
5×5 Core Fuel Mass	359 kg UO ₂
Density UO ₂	10.4 gm/cm ³
Pellet Diameter	0.423 inches
Diametrical Gap (Helium Filled)	0.0085 inches
Zircaloy-2 Cladding Thickness	0.0205 inches
Fuel Pin Outside Diameter	0.4725 inches
Fuel Pin Lattice	Rectangular
Center to Center Distance	0.606 inches × 0.524 inches
H ₂ O/UO ₂ Ratio	1.135
Length of Active Fuel	24 inches
Fuel Pins per Assembly	25
Fuel Pins in a 25 Assembly Core	625
<u>Box Assembly</u>	
Material	Zircaloy-2
Inside Dimensions	2.62 inches × 3.03 inches
Wall Thickness	0.060 inches
Clearance Between Boxes	0.040 inches
Clearance Between Control Rod Shroud and Box	0.060 inches
<u>Moderator</u>	
Material	H ₂ O (<i>light water</i>)
Temperature	70°F – 105°F

Reflector

Light water, Graphite, Beryllium

Primary Coolant / Flow Rate

De-Ionized Light water, 500 gpm

Secondary Coolant / Flow Rate

Light water, 700 gpm

Control Rods

Number

Safety Rods – 3

Pulse – 1

Absorber Material

Ag-In-Cd (80 w% - 15 w% - 5 w%)

Guide Shroud Material

Aluminum

Shape

Rectangular

Transverse Dimensions

Guide

6.34 inches × 0.44 inches

Absorber

4.85 inches × 0.18 inches

Clearance Absorber to Guide

0.0625 inches

Absorber Clad Material

Tin-Nickel

Withdrawal Speed

7.5 inches/min

Neutron Source

Pu-Be (5 Ci)